

Wave nature of particles - Numerical Problems Questions with Answers, Solution

EXAMPLE 7.6

Calculate the momentum and the de Broglie wavelength in the following cases:

- i) an electron with kinetic energy 2 eV .
- ii) a bullet of 50 g fired from rifle with a speed of 200 m/s
- iii) a 4000 kg car moving along the highways at 50 m/s

Hence show that the wave nature of matter is important at the atomic level but is not really relevant at macroscopic level.

Solution:

- i) Momentum of the electron is

$$p = \sqrt{2mK} = \sqrt{2 \times 9.1 \times 10^{-31} \times 2 \times 1.6 \times 10^{-19}} \\ = 7.63 \times 10^{-25} \text{ kg ms}^{-1}$$

Its de Broglie wavelength is

$$\lambda = \frac{h}{p} = \frac{6.626 \times 10^{-34}}{7.63 \times 10^{-25}} = 0.868 \times 10^{-9} \text{ m} \\ = 8.68 \text{ \AA}$$

- ii) Momentum of the bullet is

$$p = mv = 0.050 \times 200 = 10 \text{ kgms}^{-1}$$

Its de Broglie wavelength is

$$\lambda = \frac{h}{p} = \frac{6.626 \times 10^{-34}}{10} = 6.626 \times 10^{-35} \text{ m}$$

- iii) Momentum of the car is

$$p = mv = 4000 \times 50 = 2 \times 10^5 \text{ kgms}^{-1}$$

Its de Broglie wavelength is

$$\lambda = \frac{h}{p} = \frac{6.626 \times 10^{-34}}{2 \times 10^5} = 3.313 \times 10^{-39} \text{ m}$$

From these calculations, we notice that electron has significant value of de Broglie wavelength ($\approx 10^{-9}m$ which can be measured from diffraction studies) but bullet and car have negligibly small de Broglie wavelengths associated with them ($\approx 10^{-33}m$ and $10^{-39}m$ respectively, which are not measurable by any experiment). This implies that the wave nature of matter is important at the atomic level but it is not really relevant at the macroscopic level.

EXAMPLE 7.7

Find the de Broglie wavelength associated with an alpha particle which is accelerated through a potential difference of 400 V. Given that the mass of the proton is 1.67×10^{-27} kg.

Solution

An alpha particle contains 2 protons and 2 neutrons. Therefore, the mass M of the alpha particle is 4 times that of a proton (m_p) (or a neutron) and its charge q is twice that of a proton ($+e$).

The de Broglie wavelength associated with it is

$$\begin{aligned}\lambda &= \frac{h}{\sqrt{2MqV}} = \frac{h}{\sqrt{2 \times (4m_p) \times (2e) \times V}} \\ &= \frac{6.626 \times 10^{-34}}{\sqrt{2 \times 4 \times 1.67 \times 10^{-27} \times 2 \times 1.6 \times 10^{-19} \times 400}} \\ &= \frac{6.626 \times 10^{-34}}{4 \times 20 \times 10^{-23} \sqrt{1.67 \times 1.6}} = 0.00507 \text{ \AA}\end{aligned}$$

EXAMPLE 7.8

A proton and an electron have same de Broglie wavelength. Which of them moves faster and which possesses more kinetic energy?

Solution

We know that $\lambda = h/\sqrt{2mK}$

Since proton and electron have same de Broglie wavelength, we get

$$\frac{h}{\sqrt{2m_p K_p}} = \frac{h}{\sqrt{2m_e K_e}} \quad (\text{or}) \quad \frac{K_p}{K_e} = \frac{m_e}{m_p}$$

Since $m_e < m_p$, $K_p < K_e$, the electron has more kinetic energy than the proton.

$$\frac{K_p}{K_e} = \frac{\frac{1}{2}m_p v_p^2}{\frac{1}{2}m_e v_e^2} \quad (\text{or}) \quad \frac{v_p}{v_e} = \sqrt{\frac{K_p m_e}{K_e m_p}}$$

$$\frac{v_p}{v_e} = \sqrt{\frac{m_e^2}{m_p^2}} = \frac{m_e}{m_p} \quad \text{since} \quad \frac{K_p}{K_e} = \frac{m_e}{m_p}$$

Since $m_e < m_p$, $v_p < v_e$, the electron moves faster than the proton.

X-Rays - Numerical Problems Questions with Answers, Solution

EXAMPLE 7.9

Calculate the cut-off wavelength and cut-off frequency of x-rays from an x-ray tube of accelerating potential 20,000 V.

Solution

The cut-off wavelength of the characteristic x-rays is

$$\lambda_c = \frac{12400}{V} \text{ \AA} = \frac{12400}{20000} \text{ \AA}$$

$$= 0.62 \text{ \AA}$$

$$= 0.62 \text{ \AA}$$

The corresponding frequency is

$$\nu_0 = \frac{c}{\lambda_0} = \frac{3 \times 10^8}{0.62 \times 10^{-10}} = 4.84 \times 10^{18} \text{ Hz}$$

IV Numerical problems

1. How many photons per second emanate from a 50 mW laser of 640 nm?

[Ans: $1.61 \times 10^{17} \text{ s}^{-1}$]

Given Data:

$$\text{Power (P)} = 50 \text{ mW} = 50 \times 10^{-3} \text{ W}$$

$$\lambda = 640 \text{ nm} = 640 \times 10^{-9} \text{ m}$$

Number of photons per second (n) = ?

$$\text{The value of } hc = 6.626 \times 10^{-34} \times 3 \times 10^8$$

$$hc = 1.988 \times 10^{-25} \text{ Jm}$$

Solution:

$$\text{Energy of each photon (E)} = h\nu = \frac{hc}{\lambda}$$

$$\therefore E = \frac{1.988 \times 10^{-25}}{640 \times 10^{-9}}$$

$$E = 3.1 \times 10^{-19} \text{ J}$$

Power = Energy per second

$$P = \frac{E}{t} = nE$$

$$\therefore n = \frac{P}{E} = \frac{50 \times 10^{-3}}{3.1 \times 10^{-19}}$$

$$n = 1.612 \times 10^{17} \text{ Photons per second}$$

2. Calculate the maximum kinetic energy and maximum velocity of the photoelectrons emitted when the stopping potential is 81 V for the photoelectric emission experiment.

[[Ans: $1.3 \times 10^{-17} \text{ J}$; $5.3 \times 10^6 \text{ ms}^{-1}$]]

Given Data:

$$V_0 = 81\text{V}$$

$$K_{\max} = ? ; \quad v_{\max} = ?$$

Solution:

$$K_{\max} = eV_0$$

$$= 1.6 \times 10^{-19} \times 81$$

$$K_{\max} = 1.296 \times 10^{-17} \text{ J} \simeq 1.3 \times 10^{-17} \text{ J}$$

$$v_{\max} = 5.93 \times 10^5 \sqrt{V_0}$$

$$= 5.93 \times 10^5 \times \sqrt{81}$$

$$v_{\max} = 5.3 \times 10^6 \text{ ms}^{-1}$$

3. Calculate the energies of the photons associated with the following radiation: (i) violet light of 413 nm (ii) X-rays of 0.1 nm (iii) radio waves of 10 m.

[Ans: 3eV ; 12424eV ; 1.24×10⁻⁷ eV]

Solution:

We know that $hc = 1.988 \times 10^{-25} \text{ Jm}$

(i) For violet : $\lambda = 413 \times 10^{-9} \text{ m}$

$$E_v = \frac{hc}{\lambda} = \frac{1.988 \times 10^{-25}}{413 \times 10^{-9}}$$

$$E_v = 4.813 \times 10^{-19} \text{ J}$$

$$= \frac{4.813 \times 10^{-19}}{1.6 \times 10^{-19}} \text{ eV}$$

$$E_v = 3 \text{ eV}$$

ii) For X-rays: $\lambda = 0.1 \times 10^{-9} \text{ m}$

$$E_x = \frac{hc}{\lambda} = \frac{1.988 \times 10^{-25}}{0.1 \times 10^{-9}}$$

$$E_x = 1.988 \times 10^{-15} \text{ J}$$

$$= \frac{1.988 \times 10^{-15}}{1.602 \times 10^{-19}} \text{ eV}$$

$$= 1.242 \times 10^4 \text{ eV}$$

$$E_x = 12420$$

iii) For a Radio waves: $\lambda = 10 \text{ m}$

$$E_R = \frac{hc}{\lambda} = \frac{1.988 \times 10^{-25}}{10}$$

$$E_R = 1.988 \times 10^{-26} \text{ J}$$

$$= \frac{1.988 \times 10^{-26}}{1.6 \times 10^{-19}} \text{ eV}$$

$$E_R = 1.243 \times 10^{-7} \text{ eV}$$

4. A 150 W lamp emits light of mean wavelength of $5500 \text{ \AA}$. If the efficiency is 12%, find out the number of photons emitted by the lamp in one second.

[Ans: 4.98×10^{19}]

Given Data:

$$\text{Power (P)} = 150 \text{ W}; \quad \lambda = 5500 \text{ \AA}$$

$$\eta = 12\% = \frac{12}{100}$$

Number of photons per second (n) = ?

Solution:

$$E = \frac{hc}{\lambda} = \frac{1.988 \times 10^{-25}}{5500 \times 10^{-10}}$$

$$E = 3.611 \times 10^{-19} \text{ J}$$

$$\text{Efficiency } (\eta) = \frac{P_{out}}{P_{in}} = \frac{12}{100}$$

$$P_{out} = \frac{12}{100} \times P_{in}$$

$$P = \frac{12}{100} \times 150 = 18 \text{ W}$$

Number of photons per second

$$n = \frac{P}{E} = \frac{18}{3.611 \times 10^{-19}}$$

$$n = 4.984 \times 10^{19} \text{ Photons per second}$$

5. How many photons of frequency 10^{14} Hz will make up 19.86 J of energy?

[Ans: 3×10^{20}]

Given Data:

$$\nu = 10^{14} \text{ Hz} ; E_{\text{Tot}} = 19.86 \text{ J} ; n = ?$$

Solution:

$$E_{\text{each}} = h\nu$$

$$= 6.626 \times 10^{-34} \times 10^{14}$$

$$E_{\text{each}} = 6.626 \times 10^{-20} \text{ J}$$

$$E_{\text{Tot}} = nE_{\text{each}}$$

$$\therefore n = \frac{E_{\text{Tot}}}{E_{\text{each}}} = \frac{19.86}{6.626 \times 10^{-20}}$$

$$n = 3 \times 10^{20} \text{ Photons}$$

6. What should be the velocity of the electron so that its momentum equals that of $4000 \text{ \AA}$ wavelength photon.

[Ans: 1818 ms^{-1}]

Given Data:

$$\lambda = 4000 \text{ \AA} = 4 \times 10^{-7} \text{ m}$$

$$p = ? ; V = ?$$

Solution:

$$p = \frac{h}{\lambda} = \frac{6.62 \times 10^{-34}}{4 \times 10^{-7}}$$

$$p = 1.655 \times 10^{-27} \text{ kg ms}^{-1}$$

$$\text{Velocity of the electron is } v = \frac{p}{m} = \frac{1.655 \times 10^{-27}}{9.1 \times 10^{-31}}$$

$$v = 1818 \text{ ms}^{-1}$$

7. When a light of frequency $9 \times 10^{14} \text{ Hz}$ is incident on a metal surface, photoelectrons are emitted with a maximum speed of $8 \times 10^5 \text{ ms}^{-1}$. Determine the threshold frequency of the surface.

[Ans: $4.61 \times 10^{14} \text{ Hz}$]

Given Data:

$$\nu = 9 \times 10^{14} \text{ Hz} ; \nu_{\max} = 8 \times 10^5 \text{ ms}^{-1}; \nu_0 = ?$$

Solution:

$$\phi = h\nu - \frac{1}{2}m\nu_{\max}^2$$

$$= 6.62 \times 10^{-34} \times 9 \times 10^{14} - \frac{1}{2} \times 9.1 \times 10^{-31} \times (8 \times 10^5)^2$$

$$= 59.58 \times 10^{-20} - 29.12 \times 10^{-20}$$

$$\phi = h\nu_0 = 30.46 \times 10^{-20}$$

$$\therefore \nu_0 = \frac{30.46 \times 10^{-20}}{6.62 \times 10^{-34}}$$

$$\nu_0 = 4.601 \times 10^{14} \text{ Hz}$$

8. When a $6000 \text{ \AA}$ light falls on the cathode of a photo cell and produced photoemission. If a stopping potential of 0.8 V is required to stop emission of electron, then determine the (i) frequency of the light (ii) energy of the incident photon (iii) work function of the cathode material (iv) threshold frequency and (v) net energy of the electron after it leaves the surface.

[Ans: $5 \times 10^{14} \text{ Hz}$; 2.07 eV ; 1.27 eV ; $3.07 \times 10^{14} \text{ Hz}$; 0.8 eV]

Given Data:

$$\lambda = 6000\text{\AA} = 6 \times 10^{-7} \text{ m}$$

$$V_o = 0.8 \text{ V}$$

$$\therefore K = eV_o = 0.8 \text{ eV}$$

$$\nu = ?; E_p = ?; \phi_o = ?; \nu_o = ?; E_{net} = ?$$

Solution:

i) Frequency of the light:

$$\nu = \frac{c}{\lambda} = \frac{3 \times 10^8}{6 \times 10^{-7}}$$

$$\nu = 5 \times 10^{14} \text{ Hz}$$

ii) Energy of the photon:

$$E_p = h\nu = 6.62 \times 10^{-34} \times 5 \times 10^{14}$$

$$E_p = 3.31 \times 10^{-19} \text{ J}$$

$$= \frac{3.31 \times 10^{-19}}{1.6 \times 10^{-19}}$$

$$E_p = 2.068 \text{ eV} \approx 2.07 \text{ eV}$$

iii) Work function:

$$\phi_o = E - K$$

$$= 2.07 - 0.8$$

$$\phi_o = 1.27 \text{ eV}$$

iv) Threshold frequency:

$$\nu_o = \frac{\phi_o}{h} = \frac{1.27 \times 1.6 \times 10^{-19}}{6.62 \times 10^{-34}}$$

$$\nu_o = 3.07 \times 10^{14} \text{ Hz}$$

v) Energy of the photo electron:

$$K = h\nu - \phi_o = E - \phi_o$$

$$= 2.07 - 1.27 = \boxed{0.8 \text{ eV}}$$

9. A $3310 \text{\AA}$ photon liberates an electron from a material with energy $3 \times 10^{-19} \text{ J}$ while another $5000 \text{\AA}$ photon ejects an electron with energy $0.972 \times 10^{-19} \text{ J}$ from the same material. Determine the value of Planck's constant and the threshold wavelength of the material.

[Ans: $6.62 \times 10^{-34} \text{ Js}$; $6620 \times 10^{-10} \text{ m}$]

Given Data:

$$\lambda_1 = 3310 \text{ \AA} = 3.31 \times 10^{-7} \text{ m}$$

$$K_1 = 3 \times 10^{-19} \text{ J}$$

$$\lambda_2 = 5000 \text{ \AA} = 5 \times 10^{-7} \text{ m}$$

$$K_2 = 0.972 \times 10^{-19} \text{ J}$$

Planck's constant (h) = ? ; λ_o = ?

Solution:

$$v_1 = \frac{c}{\lambda_1} = \frac{3 \times 10^8}{3.31 \times 10^{-7}} = 9.06 \times 10^{14} \text{ Hz}$$

$$v_2 = \frac{c}{\lambda_2} = \frac{3 \times 10^8}{5 \times 10^{-7}} = 6 \times 10^{14} \text{ Hz}$$

i) For Planck's constant:

From Einstein's equation

$$hv_1 = \phi_o + K_1 \text{ ----- (1)}$$

$$hv_2 = \phi_o + K_2 \text{ ----- (2)}$$

$$(1) - (2) \Rightarrow h(v_1 - v_2) = K_1 - K_2$$

$$h = \frac{K_1 - K_2}{v_1 - v_2}$$

$$= \frac{3 \times 10^{-19} - 0.972 \times 10^{-19}}{9.06 \times 10^{14} - 6 \times 10^{14}}$$

$$h = 6.62 \times 10^{-34} \text{ Js}$$

(ii) For Threshold wavelength:

$$\phi_o = \frac{hc}{\lambda_o} = hv_1 - K_1 \text{ (from equation (1))}$$

$$\therefore \lambda_o = \frac{hc}{hv_1 - K_1}$$

$$= \frac{6.62 \times 10^{-34} \times 3 \times 10^8}{6.62 \times 10^{-34} \times 9.06 \times 10^{14} - (3 \times 10^{-19})}$$

$$= \frac{1.986 \times 10^{-25}}{6 \times 10^{-19} - 3 \times 10^{-19}} = 6620 \times 10^{-10} \text{ m}$$

$$\lambda_o = 6620 \text{ \AA}$$

10. At the given point of time, the earth receives energy from sun at $4 \text{ cal cm}^{-2} \text{ min}^{-1}$. Determine the number of photons received on the surface of the Earth per cm^2 per minute. (Given : Mean wavelength of sun light = $5500 \text{ \AA}$)

[Ans: 4.65×10^{19}]

Given Data:

$$E = 4 \text{ Cal cm}^{-2} \text{ min}^{-1} = 4 \times 4.2 \text{ cm}^{-2} \text{ min}^{-1}$$

$$\lambda = 5500 \text{ \AA} = 5.5 \times 10^{-7} \text{ m} ; \quad n = ?$$

Solution:

$$E = nh\nu = \frac{nhc}{\lambda}$$

$$\therefore n = \frac{E\lambda}{hc}$$

$$= \frac{4 \times 4.2 \times 5.5 \times 10^{-7}}{1.988 \times 10^{-25}}$$

$$= 4.65 \times 10^{19} \text{ Photons per cm}^2 \text{ per minute}$$

11. UV light of wavelength $1800 \text{ \AA}$ is incident on a lithium surface whose threshold wavelength $4965 \text{ \AA}$. Determine the maximum energy of the electron emitted.

[Ans: 4.40 eV]

Given Data:

$$\lambda = 1800\text{\AA} = 1.8 \times 10^{-7} m$$

$$\lambda_o = 4965\text{\AA} = 4.965 \times 10^{-7} m$$

The value of $hc = 1.988 \times 10^{-25} \text{ Jm}$

$$K_{\max} = ?$$

Solution:

$$K_{\max} = h(\nu - \nu_o) = hc \left(\frac{1}{\lambda} - \frac{1}{\lambda_o} \right)$$

$$K_{\max} = hc \left(\frac{\lambda_o - \lambda}{\lambda \lambda_o} \right)$$

$$= 1.988 \times 10^{-25} \frac{(4.965 - 1.8) \times 10^{-7}}{1.8 \times 4.965 \times 10^{-14}}$$

$$K_{\max} = 7.04 \times 10^{-19} \text{ J}$$

$$= \frac{7.04 \times 10^{-19}}{1.6 \times 10^{-19}} eV$$

$$K_{\max} = 4.4 \text{ eV}$$

12. Calculate the de Broglie wavelength of a proton whose kinetic energy is equal to $81.9 \times 10^{-15} \text{ J}$. (Given: mass of proton is 1836 times that of electron).

[Ans: $4 \times 10^{-14} \text{ m}$]

Given Data:

$$K = 81.9 \times 10^{-15} \text{ J} ; \lambda_p = ?$$

$$\text{Mass of proton } (m_p) = 1836 m_e$$

Solution:

$$\lambda_p = \frac{h}{\sqrt{2m_p K}}$$

$$= \frac{6.62 \times 10^{-34}}{\sqrt{2 \times 1836 \times 9.1 \times 10^{-31} \times 81.9 \times 10^{-15}}}$$

$$= \frac{6.62 \times 10^{-11}}{1.654 \times 10^3}$$

$$\lambda = 4.002 \times 10^{-14} \text{ m}$$

13. A deuteron and an alpha particle are accelerated with the same potential. Which one of the two has i) greater value of de Broglie wavelength associated with it and ii) less kinetic energy? Explain.

[Ans: $\lambda_d = 2\lambda_\alpha$ and $K_d = K_\alpha = K_d/2$]

Given Data:

The mass number of alpha particle = 4

The mass number of deuteron = 2

The charge of alpha particle = $2e$

The charge of deuteron = $1e$

$$\therefore m_{\alpha} = 2m_d$$

$$q_{\alpha} = 2q_d$$

i) The de-Broglie wavelength (λ) = $\frac{h}{\sqrt{2mqV}}$

potential is common

$$\therefore \frac{\lambda_{\alpha}}{\lambda_d} = \sqrt{\frac{m_d q_d}{m_{\alpha} q_{\alpha}}}$$

$$= \sqrt{\frac{m_d q_d}{2m_d \times 2q_d}}$$

$$\frac{\lambda_{\alpha}}{\lambda_d} = \frac{1}{2}$$

$$\therefore \lambda_d = 2\lambda_{\alpha}$$

Wavelength of deuteron is greater.

ii) Kinetic energy $K = qV$

$$\therefore \frac{K_d}{K_{\alpha}} = \frac{q_d}{q_{\alpha}} = \frac{1}{2}$$

$$K_d = \frac{K_{\alpha}}{2}$$

Hence, kinetic energy of alpha particle is less than that of deuteron.

14. An electron is accelerated through a potential difference of 81 V. What is the de Broglie wavelength associated with it? To which part of electromagnetic spectrum does this wavelength correspond?

[Ans: $\lambda = 1.36 \text{ \AA}$ and x-rays]

Given Data:

$$V = 81V ; \lambda = ?$$

Solution:

$$\begin{aligned}\lambda &= \frac{12.27 \text{ \AA}}{\sqrt{V}} \\ &= \frac{12.27}{\sqrt{81}} \times 10^{-10}\end{aligned}$$

$$\lambda = 1.363 \text{ \AA}$$

$\therefore$ Which is belongs to X-rays.

15. The ratio between the de Broglie wavelengths associated with protons, accelerated through a potential of 512 V and that of alpha particles accelerated through a potential of X volts is found to be one. Find the value of X.

[Ans: 64 V]

Given Data:

$$V_p = 512 \text{ V} ; V_\alpha = X \text{ V} \Rightarrow \frac{v_\alpha}{v_p} = \frac{X}{512}$$

$$\text{We know that, } \frac{m_\alpha}{m_p} = 4 ; \frac{q_\alpha}{q_p} = 2$$

$$\frac{\lambda_p}{\lambda_\alpha} = 1$$

Solution:

$$\text{de Broglie wavelength } \lambda = \frac{h}{\sqrt{2mqV}}$$

$$\frac{\lambda_p}{\lambda_\alpha} = \sqrt{\left(\frac{m_\alpha}{m_p}\right)\left(\frac{q_\alpha}{q_p}\right)\left(\frac{v_\alpha}{v_p}\right)}$$

$$1 = \sqrt{4 \times 2 \times \left(\frac{X}{512}\right)}$$

$$1 = \frac{X}{64}$$

$$X = 64 \text{ V}$$